

Design and Analysis of a Two-Stage CMOS P-Type Operational Amplifier

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Abstract— A two-stage CMOS operational amplifier with a PMOS input stage is designed and analyzed using the GF180MCU process. Initial sizing is obtained through hand calculations and refined using LTspice simulations. The circuit is tuned to achieve a DC output of 1.65 V. Small-signal performance is evaluated with and without compensation. A compensation capacitor is selected to introduce a dominant pole at 1 kHz. Differences between analytical and simulated results are discussed.

I. INTRODUCTION

This work presents the design of a two-stage CMOS operational amplifier using analytical modeling and simulation. The objective is to meet specified gain and biasing requirements, followed by frequency compensation to control bandwidth. The design flow includes symbolic analysis, parameter extraction, hand calculations, and simulation-based tuning.

II. SYMBOLIC ANALYSIS

The circuit under study is a two-stage CMOS operational amplifier with a PMOS differential input stage and an NMOS common-source second stage. The amplifier is biased using a PMOS current mirror.

The input signals are expressed as:

$$vI+ = VIDC + vi+ + vicm$$

$$vI- = VIDC + vi- + vicm$$

Where $vi+$ and $vi-$ represents the differential input and $vicm$ represents the common-mode input.

A. Reference Current I_{REF}

The reference current is generated using resistor R_{REF} and the diode-connected PMOS transistor MP5.

Applying Ohm's law:

$$I_{REF} = \frac{(VDD - VSG5)}{R_{REF}}$$

Since MP5 is diode-connected, it operates in saturation and sets the bias for the current mirror.

B. Bias Current I_{BIAS1}

Transistors MP4 and MP5 form a current mirror. Therefore:

$$I_{BIAS1} = \frac{(W/L)_4}{(W/L)_5} \cdot \frac{k'p_4}{k'p_5} \cdot \frac{1 + \lambda_4 V_{SD4}}{1 + \lambda_5 V_{SD5}} \cdot I_{REF}$$

Assuming matched devices and neglecting channel-length modulation:

$$I_{BIAS1} = \frac{(W/L)_4}{(W/L)_5} \cdot I_{REF}$$

C. Bias Current I_{BIAS2}

Similarly, MP3 mirrors MP5:

$$I_{BIAS2} = \frac{(W/L)_3}{(W/L)_5} \cdot \frac{k'p_4}{k'p_5} \cdot \frac{1 + \lambda_3 V_{SD3}}{1 + \lambda_5 V_{SD5}} \cdot I_{REF}$$

Assuming matched devices and neglecting channel-length modulation:

$$I_{BIAS1} = \frac{(W/L)_3}{(W/L)_5} \cdot I_{REF}$$

D. Stage 1 Differential Gain A_{d1}

The first stage is a PMOS differential pair with an NMOS current mirror load. Under small-signal operation, the differential pair converts the differential input voltage into current. For a single-ended output:

$$A_{d1} = \frac{vo_{int}}{vi_+ - vi_-} \approx \frac{-gm_p(ro_{MP2} || ro_{MN2})}{2}$$

Where:

- gm_p is the transconductance of the PMOS input transistors.
- ro_{MP2}, ro_{MN2} are the output resistances of PMOS and NMOS devices.
- The factor $\frac{1}{2}$ arises because the differential current splits equally between the two branches.

E. Stage 1 Common-Mode Gain A_{cm1}

For common-mode input, both input transistors experience the same signal. Ideally, the differential pair rejects this signal due to symmetry.

$$A_{cm1} = \frac{vo_{int}}{vicm}$$

In the ideal case: $A_{cm1} = 0$

However, in practice, the finite output resistance of the current source MP4 leads to:

$$A_{cm1} \approx \frac{-g_{m_p}(r_{o_p} || r_{o_n})}{1 + 2g_{m_p}r_{o_{MP4}}}$$

F. Stage 2 Gain A2

The second stage is a common-source NMOS amplifier with a PMOS load.

$$A2 = \frac{v_o}{v_{o_{int}}} \approx -g_{m,MN3} \cdot (r_{o,MN3} || r_{o,MP3})$$

G. Total Differential Gain $A_{d_{total}}$

The overall gain is the product of the gains of the two stages:

$$A_{d,total} = \frac{v_o}{(v_i + -v_i -)} = A_{d1} \cdot A2$$

H. Total Common-Mode Gain $A_{cm_{total}}$

Similarly:

$$A_{cm,total} = \frac{v_o}{v_{icm}} = A_{cm1} \cdot A2$$

III. TRANSISTOR PARAMETERS

The MOSFET parameters were extracted using DC simulations in LTspice based on the GF180MCU process models. Both NMOS and PMOS transistors were characterized to obtain the threshold voltage, the channel-length modulation parameter and process transconductance parameters.

A. NMOS Parameter Extraction

1) Threshold Voltage V_{th}

The threshold voltage was extracted using the square-root method. The drain current in saturation is given by:

$$I_D = \left(\frac{1}{2}\right) k_n (V_{GS} - V_{thn})^2$$

Taking the square root:

$$\sqrt{I_D} \propto (V_{GS} - V_{thn})$$

A plot of $\sqrt{I_D}$ versus V_{GS} produces a linear region. By extrapolating this region to $\sqrt{I_D}=0$, the intercept gives:

$$V_{thn} \approx 0.63 \text{ V}$$

2) Channel-Length Modulation λ_n

The parameter λ_n was extracted from the slope of the I_D vs V_{DS} curve in the saturation region:

$$\lambda_n = \left(\frac{1}{I_{D0}}\right) * \left(\frac{dI_D}{dV_{DS}}\right)$$

Using LTspice measurements at $V_{DS} \approx 3\text{V}$:

$$\lambda_n \approx 0.093 \text{ V}^{-1}$$

3) Parameter k_n

From the saturation equation: $I_D = \left(\frac{1}{2}\right) k_n (V_{GS} - V_{thn})^2$

Solving for k_n : $k_n = \frac{2I_D}{(V_{GS} - V_{thn})^2}$

Using the operating point values:

$$k_n \approx 3.34 \times 10^{-3} \frac{\text{A}}{\text{V}^2}$$

4) Process Parameter k_n'

The process transconductance parameter is:

$$k_n' = \frac{k_n}{\left(\frac{W}{L}\right)} \approx 9.35 \times 10^{-5} \frac{\text{A}}{\text{V}^2}$$

B. PMOS Parameter Extraction

1) Threshold Voltage V_{th}

The threshold voltage was extracted using the square-root method.

$$\sqrt{|I_D|} \propto (V_{SG} - |V_{thn}|)$$

Extrapolating to zero gives: $V_{thp} \approx -0.55 \text{ V}$

2) Channel-Length Modulation λ_p

From the I_D vs V_{DS} curve in the saturation:

$$\lambda_p = \left(\frac{1}{I_{D0}}\right) * \left(\frac{dI_D}{dV_{DS}}\right)$$

$$\lambda_p \approx 0.052 \text{ V}^{-1}$$

3) Parameter k_p

From the saturation equation: $I_D = \left(\frac{1}{2}\right) k_p (V_{SG} - |V_{thp}|)^2$

Solving for k_p : $k_p = \frac{2I_D}{(V_{SG} - |V_{thp}|)^2}$

Using the operating point values:

$$k_p \approx 6.91 \times 10^{-4} \frac{\text{A}}{\text{V}^2}$$

4) Process parameter k_p'

$$k_p' = \frac{k_p}{\left(\frac{W}{L}\right)} \approx 1.93 \times 10^{-5} \frac{\text{A}}{\text{V}^2}$$

C. Discussion

The extracted parameters are consistent with expected behavior for CMOS devices. The NMOS transistor exhibits a higher transconductance parameter compared to the PMOS due to higher carrier mobility. The finite values of λ indicate the presence of channel-length modulation, which results in limited output resistance and affects the gain of the amplifier stages. These extracted parameters are used in the subsequent hand calculations to estimate gain and guide transistor sizing.

IV. HAND DESIGN CALCULATIONS

The objective is to determine I_{REF} , R_{REF} , I_{BIAS1} , I_{BIAS2} , V_{IDC} , and the transistor widths such that:

$$I_{REF} = I_{BIAS1}$$

$$V_o \approx 1.65 \text{ V}$$

$$10 < |A_{d1}| < 11, \quad 10 < |A_2| < 11$$

A midpoint target of 10.5 is selected.

A. Second Stage

$$A_2 \approx -g_{m3}(r_{o_{MN3}} || r_{o_{MP3}})$$

Using: $g_{m3} = \frac{2I_{D3}}{V_{OVn}}$, $r_o \approx \frac{1}{(\lambda I_{D3})}$

Gives: $A_2 \approx -\frac{2}{(V_{OVn}(\lambda_n + \lambda_p))}$

Thus: $V_{OVn} = \frac{2}{(10.5(\lambda_n + \lambda_p))} \approx 1.31 \text{ V}$

Substituting back: $A_2 \approx -10.5 \text{ V/V}$

B. First Stage

$$A_{d1} \approx -\frac{(g_{mp}(r_{op} || r_{on}))}{2}$$

Similarly: $A_{d1} \approx -\frac{1}{(V_{OVp}(\lambda_n + \lambda_p))}$

Hence: $V_{OVp} = \frac{1}{(10.5(\lambda_n + \lambda_p))} \approx 0.657 \text{ V}$

$$A_{d1} \approx -10.5 \frac{\text{V}}{\text{V}}$$

C. Choice of Bias Currents

A practical initial choice is: $I_{REF} = I_{BIAS1} = 100 \mu A$
 Since the differential pair current splits equally under zero differential input: $I_{D1} = I_{D2} = \frac{I_{BIAS1}}{2} = 50 \mu A$
 A convenient initial choice for the second stage is: $I_{BIAS2} = 100 \mu A$
 These values provide a reasonable starting point and can later be fine-tuned in simulation.

D. Input PMOS Widths MP1 and MP2

For each PMOS input transistor:

$$I_D = \left(\frac{1}{2}\right) k_p' \left(\frac{W}{L}\right) V_{OVp}^2$$

Solving for (W/L):

$$\left(\frac{W}{L}\right)_{MP1} = \left(\frac{W}{L}\right)_{MP2} = \frac{2I_D}{(k_p' V_{OVp}^2)}$$

$$\text{Substituting: } \left(\frac{W}{L}\right)_{MP1} = \left(\frac{W}{L}\right)_{MP2} \approx 12.0$$

$$\text{Since: } L = 0.28 \mu m \text{ then } W_{MP1} = W_{MP2} \approx 3.36 \mu m$$

E. Width of MN3

For transistor MN3: $I_{D3} = I_{BIAS2} = 100 \mu A$

And $I_{D3} = I_{BIAS2} = 100 \mu A$

$$\text{Thus, } \left(\frac{W}{L}\right)_{MN3} = 2 \frac{I_{D3}}{(k_n' V_{OVn}^2)}$$

$$\text{Substituting: } \left(\frac{W}{L}\right)_{MN3} \approx 1.25$$

$$\text{Therefore, } W_{MN3} \approx 0.35 \mu m$$

F. Current Mirror Design MP4 and MP5

Part III requires: $I_{REF} = I_{BIAS1}$

The simplest way to achieve this is to choose equal mirror ratios: $\left(\frac{W}{L}\right)_{MP4} = \left(\frac{W}{L}\right)_{MP5}$

Choose a reasonable mirror overdrive: $V_{OV45} \approx 0.20 V$

$$\text{Then: } V_{SG45} \approx |V_{thp}| + V_{OV45} \approx 0.55 + 0.20 = 0.75 V$$

$$\text{Using: } I_{REF} = \left(\frac{1}{2}\right) k_p' \left(\frac{W}{L}\right) V_{OV45}^2$$

$$\text{Gives: } \left(\frac{W}{L}\right)_{MP5} = 2 \frac{I_{REF}}{(k_p' V_{OV45}^2)}$$

$$\text{Substituting: } \left(\frac{W}{L}\right)_{MP5} \approx 259$$

$$\text{Thus, } W_{MP5} \approx 72.5 \mu m$$

$$\text{Since the two widths are equal: } I_{BIAS1} \approx I_{REF} = 100 \mu A$$

G. Calculation of R_{REF}

For diode-connected MP5:

$$V_X = V_{DD} - V_{SG5} = 3.3 - 0.75 = 2.55 V$$

$$\text{Then } R_{REF} = \frac{V_X}{I_{REF}} = \frac{2.55}{100 \times 10^{-6}} = 25.5 k\Omega$$

H. Width of MP3

To make the second-stage bias current initially equal to the reference current, choose: $\left(\frac{W}{L}\right)_{MP3} = \left(\frac{W}{L}\right)_{MP5}$

$$\text{Therefore, } W_{MP3} \approx 72.5 \mu m$$

I. Choice of MN1 and MN2

The NMOS current mirror load sets the first-stage output node $v_{o_{int}}$. Since the target is $v_{o_{int}} = 1.94 V$
 A useful initial design approach is to make MN2 slightly smaller than MN1, so that the current on the right branch is slightly lower and the node voltage rises.

A practical initial choice is: $W_{MN1} \approx 4.1 \mu m$ and $W_{MN2} \approx 3.6 \mu m$

These values provide a suitable starting point and can later be refined in simulation to move $v_{o_{int}}$ toward the desired value.

J. Calculation of V_{IDC}

To keep MP2 in saturation: $V_{SD2} \geq V_{OVp}$

Since: $v_{o_{int}} \approx 1.94 V$ the minimum source voltage of MP2 is approximately $V_S \geq v_{o_{int}} + V_{OVp} \rightarrow V_S \geq 2.60 V$

Also, $V_{SG2} \approx 1.207 V$

$$\text{Therefore, } V_{IDC} = V_S - V_{SG2} = 1.40 V$$

V. GAIN CALCULATIONS

Using the design values obtained in previous parts, the small-signal parameters of the two stages are calculated first, then the stage and the total gains are determined.

A. Small-Signal Parameters

For Stage 1, each PMOS input transistor carries half of the bias current: $I_{D1} = I_{D2} = \frac{I_{BIAS1}}{2} = 50 \mu A$

$$\text{Thus, } g_{mp} = 2 \frac{I_D}{V_{OVp}} = \frac{2 \times 50 \times 10^{-6}}{0.657} \approx 1.522 \times 10^{-4} S$$

$$r_{op} = \frac{1}{(\lambda_p I_D)} = \frac{1}{(0.052 \times 50 \times 10^{-6})} \approx 384.6 k\Omega$$

$$r_{on} = \frac{1}{(\lambda_n I_D)} = \frac{1}{(0.093 \times 50 \times 10^{-6})} \approx 215.1 k\Omega$$

$$r_{op} || r_{on} \approx 137.9 k\Omega$$

For the PMOS bias transistor MP4:

$$r_{oMP4} = \frac{1}{(\lambda_p I_{BIAS1})} \approx 192.3 k\Omega$$

$$\text{For Stage 2: } g_{m3} = \frac{2I_{BIAS2}}{V_{OVn}} = \frac{2 \times 100 \times 10^{-6}}{1.31} \approx 1.527 \times 10^{-4} S$$

$$r_{oMN3} = \frac{1}{(\lambda_n I_{BIAS2})} = 107.5 k\Omega$$

$$r_{oMP3} = \frac{1}{(\lambda_p I_{BIAS2})} = 192.3 k\Omega$$

$$r_{oMN3} || r_{oMP3} \approx 68.97 k\Omega$$

B. Stage 1 Differential Gain

For the single-ended PMOS differential pair with active load:

$$A_{d1} \approx -\frac{(g_{mp}(r_{op} || r_{on}))}{2} = -10.50 V/V$$

C. Stage 1 Common- Mode Gain

Using the finite output resistance of the PMOS bias source:

$$A_{cm1} \approx -\frac{(g_{mp}(r_{op} || r_{on}))}{(1 + 2 g_{mp} r_{oMP4})} \approx -0.353 \frac{V}{V}$$

D. Stage 2 Gain

For the common-source second stage:

$$A_2 \approx -g_{m3}(r_{oMN3} || r_{oMP3})$$

$$\text{Substituting: } A_2 \approx -10.53 \frac{V}{V}$$

E. Total Differential Gain

The total differential gain is the product of the two stage gains: $A_{d, total} = A_{d1} A_2 \approx 100.52 V/V$

F. Total Common-Mode Gain

Similarly $A_{cm, total} = A_{cm1}A_2 = 3.71\text{V/V}$

VI. SIMULATION RESULTS

The circuit was simulated in LTspice using the initial values obtained in Part III. The reference current and bias voltages were already close to the desired values, so R_{REF} and V_{IDC} were kept unchanged. The main tuning step was increasing the width of MN3 from $0.35\text{ }\mu\text{m}$ to $3.3\text{ }\mu\text{m}$ in order to shift the output voltage toward mid-supply. After tuning, the output voltage became: $V_o \approx 1.655\text{V}$ which is very close to 1.65V .

Parameter	Calculated/Simulated Parameter (initial)	Simulated Parameter (final)
IREF	$100\text{ }\mu\text{A}$	$105.892\text{ }\mu\text{A}$
RREF	$25.5\text{ k}\Omega$	$25.5\text{ k}\Omega$
IBIAS1	$100\text{ }\mu\text{A}$	$118.228\text{ }\mu\text{A}$
IBIAS2	$100\text{ }\mu\text{A}$	$161.148\text{ }\mu\text{A}$
VIDC	1.40 V	1.40V
Transistor width WMP1	$3.4\text{ }\mu\text{m}$	$3.4\text{ }\mu\text{m}$
Transistor width WMP2	$3.4\text{ }\mu\text{m}$	$3.4\text{ }\mu\text{m}$
Transistor width WMP3	$72.5\text{ }\mu\text{m}$	$72.5\text{ }\mu\text{m}$
Transistor width WMP4	$72.5\text{ }\mu\text{m}$	$72.5\text{ }\mu\text{m}$
Transistor width WMP5	$72.5\text{ }\mu\text{m}$	$72.5\text{ }\mu\text{m}$
Transistor width WMN1	$4.1\text{ }\mu\text{m}$	$4.1\text{ }\mu\text{m}$
Transistor width WMN2	$3.6\text{ }\mu\text{m}$	$3.6\text{ }\mu\text{m}$
Transistor width WMN3	$0.35\text{ }\mu\text{m}$	$3.3\text{ }\mu\text{m}$

The final simulated values show that the design successfully achieved the DC output target, although the bias currents increased slightly from their initial values due to transistor non-idealities and tuning effects.

VII. SMALL-SIGNAL RESULTS WITHOUT C_C

Using the final tuned parameters from the previous part, AC simulations were performed without the compensation capacitor C_C . Differential and common-mode excitations were applied to extract the small-signal gains.

Parameter	Calculated Small- without C_C	Simulated Small-Signal Results without C_C
Stage1 differential gain Ad1	-10.5 V/V	$\approx 3.8\text{V/V}$
Stage1 common-mode gain Acml	-0.35 V/V	$\approx 0.053\text{ V/V}$

Parameter	Calculated Small- without C_C	Simulated Small-Signal Results without C_C
Stage2 gain A2	-10.5 V/V	$\approx 1.68\text{ V/V}$
Total differential gain Adtotal	110 V/V	$\approx 6.5\text{V/V}$
Total common-mode gain Acmtotal	3.7 V/V	$\approx 0.094\text{V/V}$
Bandwidth	X	$1\text{--}2\text{ kHz}$

The simulated gains are significantly lower than the calculated values, while the bandwidth is on the order of a few kHz.

VIII. SMALL-SIGNAL RESULTS WITH C_C

A compensation capacitor was connected between $v_{o_{int}}$ and v_o . By tuning its value, the selected capacitor was: $C_C = 4\text{pF}$. The value produced a dominant pole at approximately: $f_p \approx 1\text{kHz}$.

Parameter	Simulated with C_C
Ad1	3.8 V/V
Acm1	0.053 V/V
A2	1.49 V/V
Adtotal	6.5 V/V
Acmtotal	0.094 V/V
Bandwidth	1 kHz

The addition of C_C established the required dominant pole while keeping the low-frequency gains approximately unchanged.

IX. DISCUSSION

The simulated results differ from the calculated values due to non-ideal transistor behavior not accounted for in hand analysis. The analytical model assumes ideal devices with constant parameters, whereas the LTspice simulations include channel-length modulation, mobility degradation, and parasitic effects, which reduce the effective output resistance and gain.

In addition, the tuning process required increasing the width of MN3, which reduced the output resistance of the second stage and significantly lowered A_2 . This, in turn, reduced the overall gain. This highlights a trade-off between achieving the desired DC output voltage and maintaining high gain.

Finally, the compensation capacitor C_C modifies the frequency response by introducing a dominant pole around 1 kHz , improving stability without significantly affecting the DC gain.